

Sugarcane Plantation & Its Impact on Land in Brazil

A deep-dive research report — prepared for TrueTech Inc. / Agroverse network | July 2026
Trigger: Paloma raised a concern that sugarcane production may not be environmentally friendly. This report examines the evidence, separates myth from reality, and frames what it means for our sugar sourcing program (Anabel / 凯心 — 50,000 t/month for China).

1. Executive Summary

- Brazil is the **world's largest sugarcane producer** — ~668.8 MMT on ~8.9–9.1 million hectares (MY2025/26–26/27).
- The environmental record is **genuinely mixed**: historic practices (pre-harvest burning, monoculture, slave-like labor in some regions, indirect deforestation via land displacement) created real damage — but the sector has transformed dramatically since ~2007–2017.
- Modern center-south mills (where our 8 shortlisted mills operate) are now: ~99% mechanically green-harvested in São Paulo, largely Bonsucro/RenovaBio aligned, zero-deforestation committed, and net carbon-positive per liter of ethanol vs gasoline.
- The main residual risk is **policy-driven**: Decree 10,084 (2019) revoked agroecological zoning, re-opening the Amazon & Pantanal to cane expansion in principle. Pará — where Natalia's mill connection sits — is on this frontier.
- Bottom line**: "Is sugar environmentally friendly?" is the wrong question. The right question is **which mill, which practices, which certifications, which region**. That's exactly how we should frame sourcing due diligence.

2. Scale & Geography of the Industry

Metric	Value (latest)
National production MY2025/26	668.8 MMT sugarcane (Conab, ~1.2% down YoY)
Harvested area MY2025/26	~8.9 M ha → forecast ~9.1 M ha MY2026/27 (largest ever)
São Paulo share	~52% of production (343 MMT)
Goiás / Minas Gerais	12% / 11%
Center-South region	~65% of national production
North region (incl. Pará)	Only ~52,000 ha (0.6%) — tiny but +5% growing; 3.8 MMT
Alagoas / Pernambuco (N-NE)	2.5% / 2% of production

Key insight: **~99% of the cane area sits in the Center-South** (São Paulo, Goiás, Minas, Mato Grosso do Sul, Paraná). The Amazon states are a minuscule but fast-growing frontier — directly relevant to the Pará mill question.

3. Land Use & Deforestation — the Core Concern

3.1 Direct deforestation

- **ZAE Cana (Decree 6,961/2009)**: federal agroecological zoning that *excluded* the Amazon, Pantanal, indigenous lands, protected areas and *prohibited deforestation* for cane. It channeled expansion onto ~64 M ha of mostly pasture/degraded land — 7.5% of Brazil.
- **Decree 10,084 (Nov 2019, Bolsonaro)**: revoked ZAE Cana, re-opening Amazon & Pantanal in principle. Experts warn of deforestation risk and land speculation; the industry itself didn't ask for it, and it threatens EU-Mercosur trade optics.
- Studies: sugarcane expansion into ecologically marginal Cerrado landscapes **exacerbates soil degradation**; direct conversion of forests creates a carbon debt of **608 Mg CO₂/ha (Amazon)** and 142 Mg CO₂/ha (Cerrado) — decades to centuries to repay.

3.2 Indirect land-use change (ILUC) — the real, documented mechanism

- Research (2002–2012) shows sugarcane **indirectly** contributed to Amazon deforestation: cane displaced cattle/soy from São Paulo & Cerrado, which pushed ranching/soy into the Amazon frontier ("leakage").
- A São Paulo study found cane expansion *increased* cattle ranching's land-clearing impact in the Amazon — the classic rebound effect.
- This is why "zero-deforestation at the mill gate" is necessary but not sufficient — the supply-chain leakage question matters.

3.3 Pará — the frontier case

- Esalq/USP estimated Pará could host **9 M ha** of cane — a 136% boost to Brazilian ethanol output *if* it developed.
- Conab: Pará currently harvests only ~736,000 tons of cane — a rounding error nationally, but the vector for future Amazon-region expansion.
- **For our deal**: a mill in Pará carries inherently higher deforestation/ESG risk than a center-south mill. If we can source from São Paulo/Minas/Goiás instead, Paloma's concern is substantially answered.

4. Soil Impacts

Issue	Evidence	Verdict
Soil compaction	Mechanized operations reduce porosity; land-use change studies confirm compaction	Real, but manageable
Monoculture nutrient drain	Cane is a heavy NPK user — 13–17% of Brazil's fertilizer	Real; offset by fertigation
Trash retention (green harvest)	~10–15 t/ha of leaves left on soil → builds soil organic carbon, cuts erosion	Strong positive since 2007+
Vinasse fertigation	351.4 B liters/year of vinasse recycled as fertilizer/irrigation — replaces synthetic K, improves soil moisture	Net positive if managed

Conversion of degraded pasture	Studies: cane on degraded pasture <i>improves</i> soil carbon & GHG balance vs the pasture baseline	Best-case scenario
--------------------------------	---	--------------------

Summary: soil outcomes depend almost entirely on management regime. Modern green-harvest + vinasse recycling is widely regarded as soil-improving; legacy burnt-harvest monoculture was degrading.

5. Water Use

- **Sugarcane water footprint:** ~1,190 L/kg in Brazil vs ~1,600 L/kg world average — **below global average** (rain-fed in most of center-south).
- **Ethanol water footprint:** 2,245 L per liter ethanol — 2nd most efficient crop globally (sugar beet 1,176; corn 2,486; wheat 4,339; sorghum 8,317).
- **Key point:** most Brazilian cane is *rain-fed* — blue (irrigation) water is a small share. Local stress exists in some regions but is not systemic.
- Risk to watch: vinasse/agrochemical runoff into streams and groundwater if improperly applied; riparian protection is part of the SP Green Protocol.

6. Air Pollution & Pre-Harvest Burning

- **Legacy practice:** burning fields before manual harvest produced severe local air pollution — PM2.5 levels **3x higher** during harvest (101.8 vs 27.0 µg/m³), up to 56,000 silicate fibers/m³, PAH-linked cancer risk, and chronic kidney disease (CKDu) clusters among workers.
- **Transformation:** São Paulo law (2002) phased out burning over 3 decades; the **Green Protocol** (2007) + industry commitment ended nearly all burning in SP by **2017**; burning now allowed only on steep terrain until 2031.
- **Mechanization:** >99% of São Paulo harvest is now **mechanical green harvest**. Green cane also feeds cogeneration — sugarcane biomass is Brazil's 2nd-largest electricity source.
- **For sourcing:** verify the mill's green-harvest % and burning policy — modern center-south mills are clean; this is a checkable credential.

7. Agrochemicals

- Historical aerial spraying harmed nearby communities and ecosystems (documented in expansion regions, incl. Pará reports).
- Vinasse has extremely high organic load (50–150 g/L C — up to 100x domestic sewage) if discharged untreated; regulation + fertigation reuse mitigates this.
- Brazil is among the largest agrochemical consumers globally; cane's share (13–17% of NPK) is significant but its per-hectare intensity is moderate vs other crops.
- Bonsucro certification audits pesticide management — a practical filter for responsible sourcing.

8. Labor & Social Dimensions

Dimension	Detail
Historic abuses	Manual cane cutting linked to slave-like conditions, debt bondage, overexploitation (documented pre-2009, incl. a 2014 flagrant case)

National Commitment (2009)	Gov + industry + unions: 30 audited best practices for labor conditions
Employment	~730,000 jobs (RAIS 2023/24); wages among the highest in Brazilian agriculture
Mechanization trade-off	Green harvest cut manual jobs — social displacement for rural workers (real cost of the green transition)

9. Policy, Certification & Market Safeguards

- **RenovaBio (2017):** national biofuels policy; CBIO decarbonization credits; >80% of ethanol certified in 2020; now integrated with Bonsucro.
- **Bonsucro:** global standard for sustainable cane — environmental, social & economic criteria; a growing share of Brazilian mills certified.
- **EPA (2010):** Brazilian cane ethanol classified as *advanced biofuel* ($\geq 50\%$ GHG reduction vs gasoline) — an independent U.S. stamp of climate benefit.
- **UNICA / industry:** voluntary zero-deforestation commitments; Green Protocol (SP) for no-burning + riparian protection.
- **Risk:** Decree 10,084 remains in force (as of research date) — the Amazon/Pantanal safeguard is policy-dependent, so due diligence must verify each mill's actual land-use record, not just the legal default.

10. What This Means for Our Sugar Sourcing (TrueTech / Agroverse)

Framing for Paloma and for the China deal: Brazilian sugar is not categorically "dirty" or "clean" — it's mill-specific. The 8 ICUMSA-45 mills we shortlisted (Raízen, Copersucar, São Martinho, Tereos, Usina Coruripe, Atvos, Santa Adélia, Alto Alegre) are almost all center-south, modern, mechanized operators with strong ESG credentials.

- **Due-diligence checklist to attach to any mill LOI/contract:**
 1. Green (mechanical) harvest % — target $\geq 90\%$
 2. Bonsucro certification (or roadmap)
 3. RenovaBio CBIO registration
 4. Zero-deforestation commitment + no Amazon/Pantanal land holdings
 5. Vinasse/effluent management compliance
 6. Labor: National Commitment signatory; no "dirty list" (lista suja) entries
- **Prefer center-south over Pará/Amazon-frontier mills** for the China volume — same product, dramatically lower ESG risk. If we must use the Pará connection (Natalia), make certification + zero-deforestation proof a contractual condition.
- **Marketing angle:** if we can document certified sustainable sourcing, it strengthens the Agroverse "traceable, responsible" brand and answers ESG objections from buyers (China, EU, US).
- **Paloma's concern is valid as a question, answerable as a spec.** Turn it into a procurement requirement, not an argument.

11. Sources

- Conab 2025/26–26/27 crop surveys; USDA FAS Sugar Annual Brazil BR2026-0020 / BR2025-0011
- Springer: "Sustainability of sugarcane production in Brazil. A review" (2018); related LUC/CO₂ studies (2015–2021)
- Perspectives in Ecology & Conservation: Decree 10,084 / ZAE Cana analyses (2020)
- Mongabay (2019): revocation of Amazon/Pantanal cane zoning; Sugarcane Rush & Amazon coverage; local climate impacts (2013)
- Biofuelwatch: "Impacts of Expansion of Sugarcane Monocropping for Ethanol Production" (incl. Pará data)
- UNICA sector timeline & sugarcane.org (labor data, Green Protocol, mechanization)
- Bonsucro & RenovaBio integration materials; AgEcon Search (Zezza, Bonsucro EU)
- NCBI/PMC scoping review: health effects of sugarcane burning (2025); The Counter: São Paulo burn phase-out
- Embrapa / FAPESP water-footprint & vinasse data (2010); Gerbens-Leenes et al. (2009); Embrapa Piauí fertigation study
- Tandfonline: "A confirmation of the indirect impact of sugarcane on deforestation" (2017); PubMed 18536250 (Martinelli & Filoso)

Prepared by Sophia Truesight, TrueSight DAO Autopilot — July 2026. Research compiled from public sources; figures as published by cited agencies. Not investment or legal advice.